

Can a Vision–Language Model Improve the Grouping of 2D Masks into 3D Object Instances?

Samet Degirmenci
samet.degirmenci@tum.de

Florian Wissel
florian.wissel@tum.de

Kateřina Skotnicová
go98rir@tum.de

Omar Bin Faheem
faheemomar@tum.de

https://github.com/sametd04/ML3D-VLM_based_2D_mask_grouping

Technical University of Munich
Boltzmannstraße 3, 85748 Garching

Abstract

MaskClustering groups per-frame 2D instance masks into 3D object instances purely from geometric multi-view view consensus, with no notion of object semantics. We investigate whether adding semantic evidence from a vision–language model (VLM) Qwen3-VL improves this grouping, either as a direct pairwise edge scorer or fused with geometry through a lightweight MLP. On the Replica dataset, we compare five edge-scoring variants under an identical mask backbone (CropFormer): the MaskClustering baseline, zero-shot Qwen3-VL, LoRA-fine-tuned Qwen3-VL, a simple average of the two, and a learned fusion MLP over one semantic and five geometric features. On held-out test scenes (room2, office3), the fusion MLP matches or marginally exceeds the baseline (e.g. AP 0.083 vs. 0.082 on room2), while the Qwen-only scorer performs far worse (≤ 0.019 AP), and LoRA fine-tuning actively hurts generalization by over-merging objects (recall ≈ 0.97 but false-positive rate ≈ 0.89 at the default threshold). We show this failure is attributable to an extremely small fine-tuning set ($\sim 6.5k$ pairs from 4 scenes), not to the VLM signal itself. Our contribution is therefore not a positive result for semantic fusion on this dataset, but a controlled, reproducible investigation of when and why a VLM helps (or fails to help) a strong geometric baseline for 3D instance grouping.

1. Introduction

Reconstructing 3D object instances from posed RGB(-D) video is a core building block for embodied AI, scene un-

derstanding, and AR/VR. A common strategy is *bottom-up*: segment 2D instance masks in every frame with an off-the-shelf image segmenter, then group masks that belong to the same physical object across views into a single 3D instance. The recent **MaskClustering** [5] pipeline does this purely geometrically: it computes a *view-consensus* score between mask pairs from multi-view co-visibility, builds a graph over frame-level masks, and iteratively merges connected components into 3D instances. It is cheap, training-free, and surprisingly strong. But it is *semantically blind* in two opposite ways: an object seen from very few frames accumulates too little multi-view evidence to be merged confidently (*under-merging*), while two visually distinct but spatially co-located objects (e.g. a book on a table) can accumulate spuriously high view consensus and be fused into one instance (*over-merging*), in both cases because the scorer has no notion of *what* an object is.

This raises a natural question, and the one we study in this project:

Can semantic evidence from a vision–language model (VLM) improve geometric mask grouping, without requiring an expensive pairwise VLM call for every candidate pair?

Figure 1 illustrates the setting: a single RGB frame from the Replica [4] dataset with its CropFormer instance masks overlaid. Geometric view consensus is computed between masks like these across all frames of a scene; our goal is to test whether a VLM’s visual-semantic understanding of *what an object is* can add information that pure multi-view geometry misses.

We study this question through three edge scorers:



Figure 1. A Replica RGB frame with CropFormer 2D instance masks overlaid. MaskClustering groups masks like these across frames into 3D object instances using only multi-view geometric consensus.

1. **MaskClustering baseline:** Geometric view consensus only.
2. **Qwen3-VL edge scorer:** A VLM predicts $P(\text{same instance})$.
3. **Fusion MLP (ours):** A small MLP combines the Qwen probability with five geometric features into a single learned edge score.

Contributions. (1) A controlled ablation of the 2D mask backbone (CropFormer vs. SAM vs. SAM3) under MaskClustering’s native iterative clustering, identical across the three backbones. (2) A high-recall candidate-pair filter that reduces the $\mathcal{O}(N^2)$ cost of pairwise VLM scoring to a tractable subset, together with zero-shot and LoRA-fine-tuned Qwen3-VL edge scorers. (3) A lightweight fusion MLP combining one semantic and five geometric features into a single edge score, evaluated against the geometric baseline on two held-out test scenes. (4) A quantitative failure analysis explaining *why* LoRA fine-tuning of the VLM edge scorer fails to generalize, and evidence pointing to the small fine-tuning set rather than an inherent limitation of VLM-based scoring.

2. Related Work

Geometric 3D instance grouping. MaskClustering [5] is the geometric baseline our project builds on. Given a posed RGB(-D) video, it (1) segments 2D instance masks independently in each sampled frame with an off-the-shelf 2D segmenter, (2) associates masks across frames by back-projecting each mask into 3D using the depth map and camera pose and measuring how consistently pairs of masks co-occur across multiple views – the *view-consensus* score C_{view} , (3) builds an undirected graph whose nodes are per-frame masks and whose edges are mask pairs with C_{view} above a threshold, and (4) merges each connected component into a new node, repeating (3)-(4) for several rounds with progressively looser view-count thresholds until the final 3D instances are extracted. On the Replica [4] *room0*



Figure 2. 2D mask backbone comparison on the same Replica frame: CropFormer (left), SAM (middle), SAM3 (right). CropFormer produces the most instance-coherent masks; SAM fragments objects into parts; SAM3 is limited by its fixed text vocabulary.

scene this baseline achieves a class-agnostic AP of 0.149 (Table 1), which we take as our reference reproduction number before any modification. For the edge-scorer comparison in Sections 3.2 and 3.3, we simplify steps (3)-(4) to a single, non-iterative connected-components pass, applied identically across every plausible pair. Different scoring techniques were adopted and benchmarked: the original C_{view} , a VLM-derived probability on whether masks across frames belong to the same instance, and the learned fusion of the two, the details of which are shared under Section 3.

2D instance segmentation backbones. CropFormer [2] yields dense, instance-coherent masks; SAM [1] segments class-agnostically but fragments objects into parts; SAM3 (2025) adds open-vocabulary prompting, giving a semantic prior despite a fixed vocabulary and inconsistent detections.

Vision-language models for visual grouping. VLMs such as Qwen3-VL [3] can answer pairwise same-object questions, yet no prior work embeds a VLM into the edge-scoring step of a MaskClustering-style pipeline.

3. Method

3.1. Choosing the 2D Mask Backbone

Because the clustering quality is bounded by the quality of the input 2D masks, we first compare three mask generators under *identical* downstream clustering: **CropFormer** [2] (a dense, instance-coherent, task-specific segmenter), **SAM** [1] (ViT-H, class-agnostic “segment everything” with a large/stable-mask filter), and **SAM3** (text-prompted over a fixed Replica vocabulary of ~ 50 labels, confidence threshold 0.25, all-instances). Figure 2 shows example mask sets from all three on the same frames. CropFormer produces the most instance-coherent, least fragmented masks and wins the downstream AP comparison (Table 1); we therefore fix it as the mask backbone for all subsequent milestones.

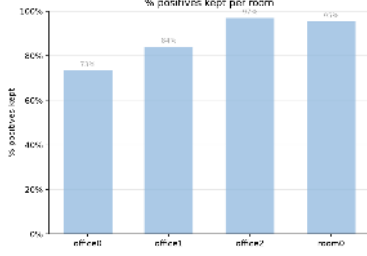


Figure 3. Percentage of positive pairs kept after candidate-pair filtering with $\tau_C = 0.3$ and $\tau = 0.05$.

3.2. Semantic Edge Scoring with Qwen3-VL

Candidate-pair filtering. Scoring every pair of masks with a VLM is $\mathcal{O}(N^2)$ per scene and infeasible at our frame counts. We therefore only query the VLM on a *high-recall* candidate set: a mask pair is retained if **any** of three cheap geometric criteria holds ($\text{DepthIoU} \geq \tau$, or $\max(O_{A \rightarrow B}, O_{B \rightarrow A}) \geq \tau$ (the maximum between the fraction of overlap of points A in B and and fraction of points B in A, respectively), or $C_{\text{view}} \geq \tau_c$; $\tau=0.05$, $\tau_c=0.30$). This OR-rule is deliberately lenient; keeping most true-positive pairs while discarding the majority of unrelated ones allows the VLM budget to be spent on plausible candidates only.

Qwen3-VL scoring. For each retained candidate pair, we render the two masks as colored outlines on their respective full RGB frames and query Qwen3-VL for $P(\text{same instance})$, i.e, the probability of two masks belonging to the same 3D instance. We evaluate two variants. The *zero-shot* variant uses the base Qwen3-VL-2B model with no task-specific training. The *LoRA fine-tuned* variant trains a low-rank adapter on the last 8 of 24 vision-encoder blocks, which capture the higher-level semantic features most relevant to instance matching. Each image’s patch tokens are combined via learned attention weights into a single global token; the two per-image tokens are concatenated and passed through a small MLP classification head (Figure 4).

3.3. Fusion MLP

Rather than relying on the VLM score alone, we train a small MLP that fuses the semantic Qwen probability p_{Qwen} with five geometric features already available from the MaskClustering pipeline: view consensus C_{view} , a log-scaled observation count $\log(1+N_{\text{obs}})$, depth IoU, and both directional overlaps $O_{A \rightarrow B}$ and $O_{B \rightarrow A}$ (Figure 5). The intuition is that an MLP score could better discriminate the masks by combining the geometric and semantic cues more optimally than simply averaging them. It is trained with a binary same-instance/different-instance target from the Replica ground-truth annotations to output a single fused

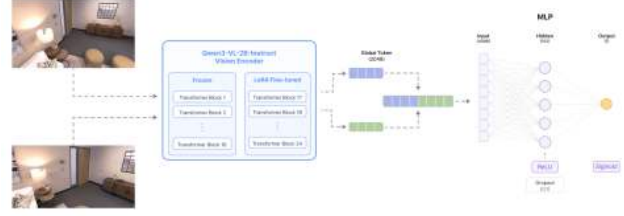


Figure 4. Qwen3-VL fine-tuning pipeline: outlined mask pairs are encoded by the vision encoder, concatenated, and passed through a small classification head to produce $P(\text{same instance})$. Unlike MaskClustering’s iterative merging, where clusters are grown step by step via repeated geometric checks, this approach scores every candidate mask pair once with Qwen3-VL’s predicted $P(\text{same instance})$ and then forms final object groups by taking connected components over the resulting pairwise graph in a single pass.

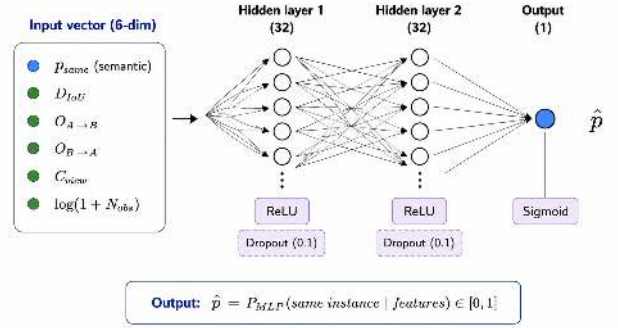


Figure 5. Fusion MLP: one semantic feature (Qwen probability) and five geometric features are concatenated and mapped to a single fused edge probability.

edge. The clustering approach is same as for the Qwen3-VL scorer. One thing to be noted is that we did not apply an iterative clustering approach for both the Qwen3-VL scorer and the Fusion MLP.

4. Experiments and Results

Setup. We use the Replica [4] dataset (8 synthetic indoor scenes with posed RGB-D frames and ground-truth 3D instance labels). The Qwen LoRA adapter is trained on *office0*, *office1*, *office2*, *room0* while the training-validation-test split for the Fusion MLP is (*room0*, *room1*, *office0*, *office1*, *office2*), validation (*office4*), and (*room2*, *office3*) respectively. This allowed for a leak-free data to test both scoring methods. All variants share the same CropFormer masks, and we report class-agnostic AP, AP@50, and AP@25.

Table 1. 2D mask backbone comparison on Replica, pooled over all 8 scenes with identical downstream clustering. CropFormer is chosen as the backbone for all later milestones.

Backbone	AP	AP@50	AP@25
CropFormer	0.123	0.260	0.472
SAM3 (conf. 0.25)	0.123	0.224	0.370
SAM (ViT-H)	0.061	0.107	0.222

4.1. Quantitative Results

Table 1 confirms CropFormer as the strongest mask backbone. Pooled over all eight Replica scenes, CropFormer and SAM3 are essentially tied in overall AP (0.123 vs. 0.123), but CropFormer is clearly better at the looser overlap thresholds (AP@50 0.260 vs. 0.224; AP@25 0.472 vs. 0.370), indicating it recovers substantially more objects at least approximately, while SAM (ViT-H) fragments objects and trails both by a wide margin. We therefore fix CropFormer as the backbone for all later milestones. Table 2 reports the edge-scorer comparison on both held-out test scenes. The Fusion MLP is the only semantic-aware variant that is competitive with, or slightly better than, the purely geometric baseline on both scenes (room2: 0.083 vs. 0.082 AP; office3: 0.127 vs. 0.128 AP, effectively tied). Simple averaging of the Qwen score with view consensus tracks the baseline closely as well. In contrast, the Qwen-only scorer, zero-shot *or* fine-tuned is far weaker on its own (AP $\leq$ 0.019 everywhere), signaling that on this dataset geometry carries almost all of the usable signal, albeit the geometric scoring was iteratively applied unlike the other scoring methods. This also suggests that the MLP Fusion method was on par with the baseline in just one iteration, hinting at the strength of combining information from both aspects. We, unfortunately, weren’t able to investigate the Fusion-MLP method (or raw semantic scoring for that matter) using an iterative approach due to computational constraints, but we hypothesize that it would drastically outscore the baseline when applied iteratively.

4.2. Qualitative Results

Figure 6 shows predicted 3D instances (one color per instance) for the geometric baseline and the Fusion MLP on zoom-ins from room2. The visually dominant differences between baseline and fusion clustering occur on harder, low-observation objects that are not highlighted here, consistent with the small quantitative gap in Table 2.

4.3. Why Does LoRA Fine-Tuning Fail to Generalize?

Fine-tuning actively *hurts* performance relative to the zero-shot model on room2 (AP 0.010 vs. 0.019). The reason is visible at the classifier level: at the default threshold

Table 2. Edge-scorer comparison on the two held-out test scenes (class-agnostic AP / AP@50 / AP@25).

room2			
Edge scorer	AP	AP@50	AP@25
View consensus (baseline)	0.082	0.189	0.423
Qwen zero-shot	0.019	0.032	0.105
Qwen fine-tuned (LoRA)	0.010	0.010	0.091
Average (Qwen + consensus)	0.075	0.191	0.430
Fusion MLP (geo + Qwen)	0.083	0.205	0.441

office3			
Edge scorer	AP	AP@50	AP@25
View consensus (baseline)	0.128	0.265	0.432
Qwen zero-shot	0.013	0.046	0.155
Qwen fine-tuned (LoRA)	0.016	0.056	0.113
Average (Qwen + consensus)	0.128	0.265	0.432
Fusion MLP (geo + Qwen)	0.127	0.262	0.427

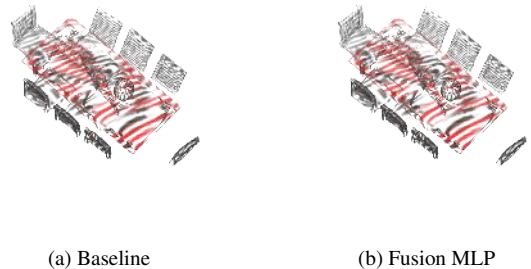


Figure 6. Zoom-in on a dining table (room2): both the geometric baseline and the Fusion MLP correctly group it into a single 3D instance.

($\tau=0.5$), the fine-tuned model reaches recall $\approx$ 0.97 but a false-positive rate of $\approx$ 0.89, essentially predicting “same instance” for almost every pair. Since clustering relies on connected components, even a few erroneous positive edges can transitively fuse many unrelated masks into one mega-cluster, devastating AP despite only mild miscalibration per prediction. This is consistent with the poorly separated positive/negative score distributions after fine-tuning (mean 0.673 vs. 0.624): raising the threshold to $\tau=0.68$ partially recovers performance but still trails the geometric baseline. We attribute this to the small fine-tuning set (6,572 pairs, 3,814 positive / 2,758 negative, four scenes, two epochs) rather than a fundamental limitation of using a VLM as an edge scorer.

5. Conclusion

We tested whether a vision–language model can improve geometric 3D mask grouping, either as a direct edge scorer

or fused with geometry. On Replica, the result is nuanced. Fusing semantic and geometric features matches or marginally exceeds using geometric cues alone iteratively in just one iteration (up to +0.001 AP). The VLM edge scorer is far weaker on its own, and naive LoRA fine-tuning makes it worse, actively hurting held-out performance through over-merging (most clearly on room2). Our failure analysis (Section 4.3), however, traces this degradation to an insufficient fine-tuning set rather than an inherent ceiling on VLM-based scoring. We conclude that there is potential in combining geometric and semantic features iteratively whilst at the same time acknowledging that the solution requires more data to derive useful semantic information from fine-tuned VLM models.

Limitations. Our method has three main shortcomings. First, single-pass clustering has no mechanism to correct a wrong pairwise prediction: one erroneous “same instance” edge can merge unrelated masks into an oversized cluster, and unlike iterative methods, our approach cannot recheck consistency as clusters grow. Second, candidate-pair filtering necessarily discards some true positive pairs to keep VLM querying tractable, capping the achievable recall regardless of scorer quality. Third, our evaluation is narrow in scope: we validate on a single small synthetic dataset with only one fine-tuning recipe, and Replica’s smooth, dense camera trajectories make the geometric baseline unusually strong, likely understating the benefit of semantic evidence on harder, real-world captures. We expand on each of these limitations, and discuss directions for future work, in Appendix B.

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A. Additional Sweep Results

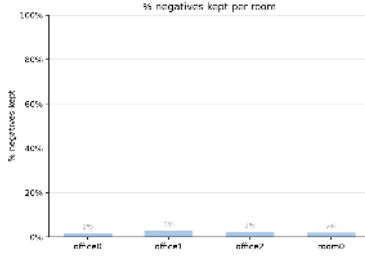


Figure 7. Percentage of negative pairs kept after candidate-pair filtering, per Replica scene, at $\tau = 0.05$ and $\tau_c = 0.3$ (the fixed thresholds also used for Figure 3’s positive-pair counterpart). Lower is better: a kept negative is a non-matching mask pair that still has to be scored by the VLM, so this bar chart quantifies the filter’s false-positive cost – the price paid, in wasted VLM queries, for the high recall shown in Figure 3.

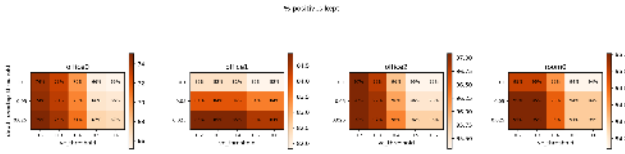


Figure 8. Percentage of ground-truth positive pairs kept as a joint function of the depth-overlap threshold τ (rows) and the view-consensus threshold τ_c (columns), shown separately for each of the 8 Replica scenes. Each panel is colored on its own min/max range, since the scenes differ substantially in scale. This complements the one-threshold-at-a-time line sweeps in the main text by showing how the two thresholds jointly trade off recall, rather than assuming their effects are independent.



(a) MaskClustering baseline.

(b) Semantic-geometric fusion.

Figure 9. Qualitative comparison of the predicted 3D object instances. Each color represents one predicted instance. The similar scene decompositions are consistent with the comparable quantitative performance of MaskClustering and the Fusion MLP, although minor differences in instance grouping remain visible.

B. Limitations & Future work

Filtering and clustering. Candidate-pair filtering always discards some positive pairs, hurting performance, but is

necessary to keep the Qwen3VL-based approach feasible. Clusters also form in a single pass over predicted “same instance” pairs, so one wrong prediction can chain unrelated masks together with no correction, leaving clusters oversized. An iterative approach like MaskClustering’s could catch such mistakes by rechecking consistency as clusters grow, but that requires repeated re-rendering and re-querying of the VLM, which was too costly here.

Scope. Our conclusions rest on one small synthetic dataset (8 Replica scenes, 2 held-out test scenes) and one LoRA recipe (vision-encoder-only adaptation with a small classification head) trained on only $\sim 6.5k$ pairs. Replica’s smooth, dense camera trajectories also make its geometric baseline unusually strong, likely understating the value of semantic evidence on sparser or noisier real-world captures.

Future work. Promising directions include: (1) an iterative, computationally efficient variant of our scoring approach that can correct clustering mistakes as they arise; (2) measuring candidate-pair recall as a function of VLM calls to quantify the recall/cost trade-off of the OR-rule filter; (3) replacing the generative pairwise VLM call with cheaper alternatives such as frozen visual-encoder embedding similarity, which our probing suggests retains much more of the encoder’s semantic signal than the current generative scoring head; and (4) testing robustness with larger VLMs and, especially, more fine-tuning data and scene diversity, the main bottleneck behind the current negative fine-tuning result